

Coursework: Project: “Design of an experimental project in Materials Chemistry and COSHH-based risk assessment”

The coursework for this module consists of designing an experimental project in Materials Chemistry. This will be carried out in groups.

Your aim is to develop a material for specific purposes, provided below for your group. You will have to research the main application, research material alternatives, available reagents, and carry out a full COSHH-based risk assessment.

Submission date:

1st of December midnight China time: Submission of the group report on QMPLUS: The report will constitute 2 pdf files. One entitled “MX Design of an experimental project in Materials Chemistry”, and the other, “MX COSHH-based risk assessment” (X will be your group number). **If you fail to comply with these instructions marks will be deducted.**

Topic:

All groups will have to develop a material in 2D format onto a specific surface for specific applications that can be shaped as desired using MEMS techniques. *Normally*, the material selected will have specific hydrophobic/hydrophilic properties, which require modification for a hypothetical specific application. Your task is to propose a material and how you will shape it in the format required, followed by a chemical modification to achieve the desired properties (essentially maintain the majority of the material, but modify the surface through a chemical modification, which would change the material interaction water). The chemistry that you use should, wherever possible, be similar to the chemistry you learned in the whole of this module or in Engineering Chemistry (if in doubt, please ask).

The starting materials and reagents you need to use **must** be commercially available (this means you are not allowed to include synthesis of photoresists or reagents in your proposal). Any articles you cite **must** be written in English. If you fail to comply with these instructions marks will be deducted.

Group	Material
M1	Negative photoresist onto silicon wafers. The shape should be a repetition of a square shape of 300 µm side repeated along the wafer, with intervals of 300 µm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain functional groups, covalently attached to the surface that enables further attachment onto the material surface using amide chemistry. The exposed substrate areas (silicon) should be hydrophobic without any reactive functional groups.
M2	Negative photoresist onto silicon wafers. The shape should be a repetition of a square shape of 300 µm side repeated along the wafer, with intervals of 300 µm side negatives. The surface of the photoresist shape should be as hydrophobic as possible and contain functional groups, covalently attached to the surface that enables further attachment onto the material surface using thiol groups. The exposed substrate areas (silicon) should be hydrophilic without any reactive functional groups.
M3	Negative photoresist onto silicon wafers. The shape should be a repetition of a square shape of 300 µm side repeated along the wafer, with intervals of 300 µm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain no reactive functional groups. The exposed substrate areas (silicon) should be hydrophobic without any reactive functional groups.
M4	Negative photoresist onto silicon wafers. The shape should be a repetition of a square shape of 300 µm side repeated along the wafer, with intervals of 300 µm side negatives. The surface of

	the photoresist shape should be as hydrophobic as possible and contain no reactive functional groups. The exposed substrate areas (silicon) should be hydrophilic without any reactive functional groups.
M5	Positive photoresist onto silicon wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain functional groups, covalently attached to the surface that enables further attachment onto the material surface using amide chemistry. The exposed substrate areas (silicon) should be hydrophobic without any reactive functional groups.
M6	Positive photoresist onto silicon wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophobic as possible and contain functional groups, covalently attached to the surface that enables further attachment onto the material surface using thiol groups. The exposed substrate areas (silicon) should be hydrophilic without any reactive functional groups.
M7	Positive photoresist onto silicon wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain no reactive functional groups. The exposed substrate areas (silicon) should be hydrophobic without any reactive functional groups.
M8	Positive photoresist onto silicon wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophobic as possible and contain no reactive functional groups. The exposed substrate areas (silicon) should be hydrophilic with reactive functional groups that enable further attachment onto the material surface using amide chemistry.
M9	Negative photoresist onto glass wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain functional groups, covalently attached to the surface that enables further attachment onto the material surface using amide chemistry. The exposed substrate areas (glass) should be hydrophobic without any reactive functional groups.
M10	Negative photoresist onto glass wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophobic as possible and contain functional groups, covalently attached to the surface that enables further attachment onto the material surface using thiol groups. The exposed substrate areas (glass) should be hydrophilic without any reactive functional groups.
M11	Negative photoresist onto glass wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain no reactive functional groups. The exposed substrate areas (glass) should be hydrophobic without any reactive functional groups.
M12	Negative photoresist onto glass wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophobic as possible and contain no reactive functional groups. The exposed substrate areas (glass) should be hydrophilic without any reactive functional groups.
M13	Positive photoresist onto glass wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain functional groups, covalently attached to the surface that enables further attachment onto the material surface using amide chemistry. The exposed substrate areas (glass) should be hydrophobic without any reactive functional groups.

M14	Positive photoresist onto glass wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophobic as possible and contain functional groups, covalently attached to the surface that enables further attachment onto the material surface using thiol groups. The exposed substrate areas (glass) should be hydrophilic without any reactive functional groups.
M15	Positive photoresist onto glass wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain no reactive functional groups. The exposed substrate areas (glass) should be hydrophobic without any reactive functional groups.
M16	Positive photoresist onto glass wafers. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophobic as possible and contain no reactive functional groups. The exposed substrate areas (glass) should be hydrophilic without any reactive functional groups.
M17	Negative photoresist onto a gold surface. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain no reactive functional groups. The exposed substrate areas (gold) should be hydrophobic without any reactive functional groups.
M18	Negative photoresist onto a gold surface. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophobic as possible and contain no reactive functional groups. The exposed substrate areas (gold) should be hydrophilic without any reactive functional groups.
M19	Positive photoresist onto a gold surface. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain functional groups, covalently attached to the surface that enables further attachment onto the material surface using amide chemistry. The exposed substrate areas (gold) should be hydrophobic without any reactive functional groups.
M20	Positive photoresist a gold surface. The shape should be a repetition of a square shape of 300 μm side repeated along the wafer, with intervals of 300 μm side negatives. The surface of the photoresist shape should be as hydrophilic as possible and contain no reactive functional groups. The exposed substrate areas (gold) should be hydrophobic without any reactive functional groups.

Reports instructions:

Report 1: "MX Design of an experimental project in Materials Chemistry"

- **Include names** of all members of the group

- **Introduction:** Not more than 1 page. Include: the polymer structure and name. Include a rationale of why you selected the material you will be working with. Include a rationale of why you selected the modification strategy you will follow.

- **Chemistry and photolithography schemes:** Not more than 1 page. Show the photolithography scheme and the chemical reactions for all the steps. Marks will be deducted if the structures are not written correctly. Use condensed structures.

- **Mechanisms:** Not more than 2 pages. Show the reaction mechanisms of all reactions you proposed.

- **References:** Include references that demonstrate the polymer is commercially available and that the chemistry can be adapted to the system of your choice.

Report 2: "MX COSHH-based risk assessment"

Complete a COSHH risk assessment form for your proposed experiments and submit it as a pdf file.